
Phase-Conditioned Student-t Mixture Regression for X-Trans Demosaicing

Method, reproducible corpus, native
implementation, and experimental limitations

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We describe an experimental X-Trans demosaicer based on phase-conditioned Student-t mixture regression (TGMR). Each of eighteen observation contracts predicts the two missing center colors from the 49 measured samples in a 7 by 7 neighborhood. A common, color-balanced observable DC is removed before regression. Fixed-degree-of-freedom Student-t fitting, tempered mixture weights, and a nine-sample shortlist reduce brittle component selection and permit a compact CPU implementation. The released configuration uses 32 components per phase and evaluates eight full experts per pixel. We distinguish retrospective BSDS development evidence from a new, reproducible corpus of 5,000 rights-reviewed images and its separately trained production-v1 candidate. On 64,000 frozen coordinates from 500 diagnostic sources, that candidate obtains 34.064 dB versus 32.316 dB for Markestijn three-pass, measured over all three center RGB channels. Average and aggregate tail errors improve, but 65 sources lose more than 0.5 dB and the worst paired loss is 6.170 dB. Subsequent synthetic hard-case replacement did not repair these failures sufficiently. The contribution is therefore a reproducible statistical demosaicing system and a documented quality-cost tradeoff, not a claim of universal superiority or completed release qualification. The corpus is publicly downloadable; model-license finalization, broader native-platform testing, and independent confirmation remain separate tasks.

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1 Introduction and scope

Demosaicing estimates the two colors that a color filter array (CFA) does not measure at each sensor location. The inverse problem is underdetermined: different color images can produce the same scalar mosaic. Local geometry, inter-channel correlation, and learned image statistics provide useful priors, but do not remove this ambiguity.

Fujifilm’s X-Trans array has a 6 by 6 repeating cell rather than Bayer’s 2 by 2 cell. Its distinct local sampling patterns require explicit treatment in a phase-dependent predictor. Dedicated frequency-domain work identifies the associated luminance/chrominance replicas [1]. RawTherapee’s Markesteyn implementation supplies a strong practical comparison; we cite the source rather than attributing an unidentified paper to that algorithm [2].

This work develops a learned, non-neural local regressor. We use TGMR for Student-t mixture regression, continuing terminology from an earlier Gaussian-mixture-regression prototype. The final components are Student-t distributions, not Gaussians. Each component supplies an affine predictor; the input-dependent mixture weights make their combination nonlinear.

The engineering and experimental contributions are:

1. A task-reduced joint model of 49 measured values and two unmeasured center colors, separately fitted for eighteen X-Trans observation patterns.
2. Fixed- ν Student-t fitting and tempered, coarse-to-fine component weighting, with a frozen K32/S9/q8 inference contract.
3. A deterministic source and patch selection recipe, commercially reusable source-license filtering, and a downloadable authenticated patch corpus.
4. Algebraically specialized native inference, same-component SIMD batching, bounded worker storage, and integration with explicit runtime fallback.
5. An evidence-linked account of both average gains and substantial failures, including an unsuccessful hard-case retraining experiment.

None of Student-t mixtures, conditional affine regression, CFA-specific predictors, or SIMD batching is claimed to be a new mathematical invention. The contribution is their particular task formulation, combination, implementation, and evaluation. We do not establish priority over every possible combination in the literature or claim patent novelty.

This revision replaces an earlier research-only account. The historical manuscript remains in Git. It distinguishes two model populations throughout: the BSDS development models and the later production-v1 candidate. Here, “production-v1” names an artifact, not a declaration that all quality and release gates passed.

2 Relation to previous work

2.1 Linear statistics, residual interpolation, and learned priors

Joint spatial-chromatic LMMSE predicts missing colors using covariance learned from complete image neighborhoods [3]. A single joint Gaussian conditioned on its observed coordinates is the affine version of this idea. A mixture can represent several local statistical populations, but must infer which populations are plausible from the incomplete CFA observation.

Residual interpolation instead reconstructs a guide and interpolates its sampled residual [4]. The corrected MLRI implementation in our development sequence is a comparison, not part of TGMR inference. TGMR uses no preliminary demosaic or guide-image target estimate.

Learned convolutional demosaicers also use population information [5], [6]. We make no claim that this compact local statistical model outperforms contemporary neural methods. Its intended advantages are an inspectable inference rule, small fixed model, and a portable CPU implementation without a neural-network runtime.

2.2 What is, and is not, reproduced from JCS-GMM

Sandeep and Jacob’s JCS-GMM learns a joint RGB patch distribution, selects a component from its observed CFA marginal, and estimates missing values from inter-channel covariance [7]. The published Bayer method uses 6 by 6 RGB patches, 150 full-covariance components, hard component selection, overlapping aggregation, and image-adaptive updates.

Our earlier full-RGB X-Trans GMM experiment isolated the local conditional estimator. It is not a numerical reproduction of that complete published algorithm. In particular, a local center-only, soft-mixture baseline without image adaptation cannot support a claim to have beaten the published JCS-GMM system. Moving from that baseline to phase-conditioned GMR changed dimension, component count, and model organization together; it was a design sequence, not a controlled dimension-only ablation.

EPLL applies an externally learned patch mixture through a whole-image restoration objective [8]. TGMR does not optimize EPLL, average overlapping reconstructed RGB patches, search nonlocal neighbors, or refine the image iteratively. It predicts one center RGB vector at a time.

2.3 Heavy tails and the inference approximation

Peel and McLachlan give a latent-scale formulation for fitting mixtures of Student-t distributions [9]. We use that established construction with fixed degrees of freedom. The conditional location of a multivariate Student-t is affine in the observed coordinates [10].

However, the released predictor is not the exact conditional-MMSE estimator of the fitted mixture: it regularizes observed scale blocks, applies a temperature of four, and renormalizes over a shortlist. We distinguish these modeling approximations from later algebraic optimizations that preserve the chosen reduced predictor.

3 Observation contract

3.1 CFA phase and target coordinates

Let $c_p(i)$ be the measured color at position i in a 7 by 7 patch with phase p . Given complete training RGB x , gather $y_i = x_{i, c_p(i)}$ in row-major spatial order. Thus $y \in \mathbb{R}^{49}$ contains only physically observable scalar samples. The two-dimensional target t contains the missing center colors in ascending RGB order:

Measured center	Target order
R	G, B
G	R, B
B	R, G

The canonical training matrix is

$$\begin{bmatrix} G & R & G & G & B & G \\ B & G & B & R & G & R \\ G & R & G & G & B & G \\ G & B & G & G & R & G \\ R & G & R & B & G & B \\ G & B & G & G & R & G \end{bmatrix}.$$

Enumerate origin rows 0 through 5, then origin columns 0 through 5, retaining the first occurrence of each distinct translated 6 by 6 matrix. This yields eighteen explicit observation/target contracts. Runtime matching uses the actual camera CFA pattern, including its phase, rather than only the center color. Supported rotated/reflected representations must match one of those contracts; they do not cause RGB channel swaps. An unsupported pattern fails before the result can be accepted.

The eighteen phase versions of a training patch are not eighteen independent photographs or patches. They are deterministic views of the same underlying RGB data. We report independent source and patch counts separately from phase-expanded training or evaluation counts.

3.2 Common observable DC

For each observed color, let $I_{p,c} = \{i : c_p(i) = c\}$. Define

$$d_p(y) = \frac{1}{3} \sum_{c \in \{R, G, B\}} \frac{1}{|I_{p,c}|} \sum_{i \in I_{p,c}} y_i.$$

Use $\tilde{y} = y - d_p(y)\mathbf{1}$ and $\tilde{t} = t - d_p(y)\mathbf{1}$. This equal-weights the three observed color means rather than over-weighting the more frequent green samples. The same full 49-sample DC is used by the coarse and full stages. No ground-truth color or initializer enters this calculation at inference.

The joint training vector is $z = [\tilde{y}^\top, \tilde{t}^\top]^\top \in \mathbb{R}^{51}$. Removing DC imposes a linear constraint on these features, making scale regularization important rather than optional.

4 Fitting and inference

4.1 Fixed-degree-of-freedom Student-t fitting

For each phase independently, fit

$$p(z) = \sum_{k=1}^K \pi_k \text{St}_\nu(z; \mu_k, \Sigma_k), \quad K = 32, \quad \nu = 3.$$

Here Σ_k denotes a scale matrix. It is not the marginal covariance: when $\nu > 2$, the latter is $\nu \Sigma_k / (\nu - 2)$. The production trainer starts with deterministic k-means++ and ten full-covariance Gaussian EM iterations, then performs thirty fixed- ν Student-t iterations in float64. The research prototype used Python and scikit-learn initialization [11]; the production path is a standalone C++ trainer.

For training sample i , compute the untempered joint responsibility $r_{ik} = p(k \mid z_i)$, using log-sum-exp, and

$$\delta_{ik} = (z_i - \mu_k)^\top \Sigma_k^{-1} (z_i - \mu_k), \quad u_{ik} = \frac{\nu + 51}{\nu + \delta_{ik}}.$$

Writing $N_k = \sum_i r_{ik}$ and $U_k = \sum_i r_{ik} u_{ik}$, the updates are

$$\begin{aligned} \pi_k^+ &= \frac{N_k}{N}, \quad \mu_k^+ = \frac{\sum_i r_{ik} u_{ik} z_i}{U_k}, \\ \Sigma_k^+ &= \frac{\sum_i r_{ik} u_{ik} (z_i - \mu_k^+)(z_i - \mu_k^+)^\top}{N_k} + \epsilon I, \quad \epsilon = 10^{-6}. \end{aligned}$$

In particular, the scale denominator is N_k , not U_k . These are the fixed-degree-of-freedom latent-scale updates [9]. Factorizations, regularization, deterministic component ordering, and finite-value checks are implementation requirements; regularization should not be mistaken for a proof of strict likelihood monotonicity in finite arithmetic. The implementation uses Cholesky factorization and triangular solves rather than forming matrix inverses explicitly; the inverse notation specifies the mathematics.

4.2 Conditional component locations

Partition each joint component into observed and target blocks. The exported predictor uses

$$S_k = \Sigma_{yy,k} + \tau^2 I, \quad \tau = 0.0003,$$

$$m_k(\tilde{y}) = \mu_{t,k} + \Sigma_{ty,k} S_k^{-1} (\tilde{y} - \mu_{y,k}).$$

Without this added regularizer, the expression is the Student-t conditional location [10]. Its conditional scale depends on the observation, but that scale is not needed to compute the location. Here τ is a model-mismatch/numerical regularizer, not a fitted sensor-noise standard deviation. Adding $\tau^2 I$ is not claimed to implement the exact convolution of a Student-t density with independent Gaussian sensor noise.

With $v_k = \tilde{y} - \mu_{y,k}$, the observed log score is

$$\ell_k = \log \pi_k - \frac{1}{2} \log |S_k| - \frac{\nu + 49}{2} \log \left(1 + \frac{v_k^\top S_k^{-1} v_k}{\nu} \right),$$

up to a phase-common constant. A dense tempered mixture would use

$$\gamma_k = \frac{\exp(\ell_k/T)}{\sum_j \exp(\ell_j/T)}, \quad T = 4.$$

Because $T \neq 1$, these are tempered weights, not ordinary Bayesian posterior probabilities under the fitted mixture. Consequently, neither their entropy nor their concentration is certified as calibrated confidence.

4.3 K32/S9/q8: what is selected

Every query uses its fixed 7 by 7 neighborhood. There is no adaptive spatial patch search. S9 denotes the central 3 by 3 marginal of the centered observation. Score all 32 components on those nine scalar samples, retain the eight highest-scoring IDs, and break ties by lower component ID. Call the shortlist Q .

Evaluate full 49-dimensional scores and affine predictions only for $k \in Q$. The final missing colors are

$$\hat{t} = d_p(y) \mathbf{1} + \sum_{k \in Q} \frac{\exp(\ell_k/T)}{\sum_{j \in Q} \exp(\ell_j/T)} m_k(\tilde{y}).$$

Copy the measured center component directly from the input. The inferred components remain signed finite values; the engine adds no sample-dependent clipping, denoising, or sharpening to this predictor.

For each output pixel:

match the actual CFA to a stored phase contract
gather 49 measured samples, reflecting only at the image boundary
compute the color-balanced common DC and subtract it
score 32 nine-sample marginals; select eight IDs stably
evaluate their full scores and two-color affine predictors
normalize the eight temperature-four weights; combine; add DC
copy the measured center color exactly

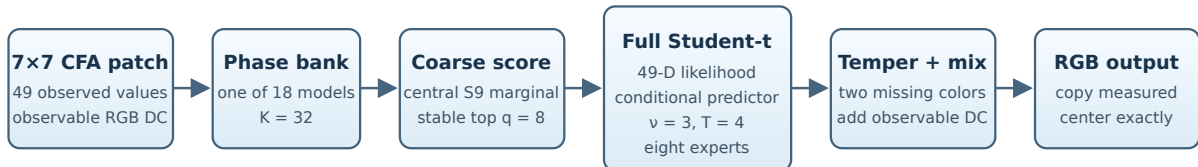


Figure 1: The inference query has a fixed spatial support. Phase lookup and component shortlisting select predictors, not other image patches.

K=32 models were fitted as K=32 mixtures, not made by deleting half the components of a K=64 fit. Shortlisting is nevertheless an approximation to dense K=32 inference, and the reduced model is not identical to dense K=64.

5 Corpus construction and training-patch selection

5.1 Development data versus the released corpus

The historical proof used 200 BSDS training images and 512 patches per image: 102,400 independent RGB patches, reused across all eighteen phase contracts. Validation and test comparisons each used twenty sources with 24 by 24 center grids, or 11,520 centers. BSDS supports the development analysis [12], but it is not the source of the bundled production-v1 coefficients. Its research-oriented distribution terms motivated a separate corpus.

The tgmrcorpus-v1 release instead contains patches derived from 5,000 rights-reviewed images [13]. Its source allocation is:

Catalog	Training	Validation	Diagnostic test	Total
Open Images	3,200	400	400	4,000
Wikimedia Commons	480	60	60	600
Smithsonian Open Access	320	40	40	400
Total	4,000	500	500	5,000

These are catalog sources, not guarantees of five thousand unprocessed camera photographs. In particular, the supplementary collections include artwork and objects.

Catalog quotas diversify image characteristics; they do not establish that the resulting distribution matches all camera RAWs.

Accepted catalog records have explicit CC BY or CC0 terms and complete attribution under the project’s documented review policy. Author grouping, a five-image author cap, exact/pixel/perceptual duplicate checks, and human people/ambiguous-duplicate review precede selection. Images must decode with a usable color interpretation and satisfy the frozen resolution gate (shortest side at least 512 pixels and at least 0.75 megapixels). Source groups do not cross splits. The release records all individual source terms; collection-level terms do not replace them. This describes the performed metadata-based review, not a legal guarantee about every underlying work.

5.2 Source balancing and nested training order

The classifier measures luminance, chroma, texture, saturation, clipping, contrast, and related quality indicators. Selection balances measurable signal strata while retaining content guardrails. Human review is not replaced by an image classifier where people safety or duplicate ambiguity requires a decision.

For learning curves, a deterministic nested order preserves the catalog proportions at 250, 500, 1,000, 2,000, and 4,000 training sources. Within catalogs, brightness/chroma/texture cells are interleaved. This avoids a prefix consisting only of Open Images. The ordering changes neither source membership nor review decisions. Validation and test sources are unchanged across these fits.

5.3 Exact coordinate-selection algorithm

There are 256 distinct 7 by 7 patches per training source and 128 per validation/test source. Coordinates denote patch top-left corners. The requested count is n . The candidate space contains $(W - 6)(H - 6)$ valid positions in the decoded linear image; distinct coordinates may still have overlapping spatial support.

The algorithm in `patch_selection.cc` is deterministic for a fixed decoded image and source seed:

1. Select 75% of the requested coordinates by seeded xorshift64* proposals modulo the number of valid positions, discarding repeated coordinates. These are approximately uniform spatial samples; modulo mapping has a negligible finite-generator bias and is not stratified sampling.
2. Form a separate unused-coordinate pool of up to $\max(4096, 32n)$ candidates, bounded by the available positions.
3. Calculate each candidate’s seven local statistics, described below.
4. Fill the remaining 25% by cycling through seventeen coverage scores, selecting the unused candidate with the highest score. Break equal scores by the lower flattened coordinate. If the pool is exhausted, use the first remaining valid coordinate.

For statistics only, RGB values are bounded to $[0, 1]$. Per-pixel luminance is $Y = 0.2126R + 0.7152G + 0.0722B$; chroma is $\max(R, G, B) - \min(R, G, B)$; saturation is chroma divided by the maximum when it is nonzero. Average these quantities over the 49 samples. Contrast is mean absolute luminance deviation from the patch mean. Texture is the root mean square of the 84 horizontal and vertical adjacent luminance differences. Clipping counts pixels whose

maximum channel is within one uint16 step of zero or one. Hue is a saturation-weighted circular mean.

The seventeen scores seek low/high luminance, low/high chroma, low/high texture, high contrast, high clipping, high saturation, and eight evenly spaced hue sectors. A hue-sector score is saturation times the cosine of the angular distance to that sector's midpoint. Thus this is deliberate signal-coverage enrichment, not selection by a demosaicer's error or by ground-truth knowledge of which method would win.

Select coordinates without replacement:

192 spatial + 64 coverage for each training source

96 spatial + 32 coverage for each evaluation source

Freeze coordinates and augmentation identities in the source manifest.

Extract RGB once; form eighteen CFA observation contracts during fitting.

The coverage portion biases the corpus toward specified signal extremes. This is intentional and applies to evaluation too; reported pooled PSNR describes this frozen sampling recipe, not a uniform average over all pixels of all ordinary photographs. Required low/middle/high brightness, chroma, and texture counts are checked independently after packing. The released low/middle/high boundaries are 0.08/0.65 for mean luminance, 0.03/0.15 for mean chroma, and 0.01/0.05 for gradient RMS. Every such marginal stratum contains at least 10,000 training and 1,000 records in each evaluation split. These are marginal-coverage guarantees, not a claim that all 27 joint cells are equally populated.

5.4 Color interpretation, augmentation, and storage

Image decoding and color management precede patch extraction. The frozen recipe starts from linearized RGB derived from rendered source images, not paired sensor RAW and spectral ground truth. Exactly every fourth patch is identity-transformed. For the remaining 75%, a source-hash/sequence rule chooses an exposure from $\{-2, -1.5, -1, -0.5, 0.5, 1, 1.5, 2\}$ stops, one of six fixed white-balance gain triplets, and a camera matrix. Gains span 0.5–2 before their Q12 quantization; four training matrix IDs and two held-out evaluation matrix IDs are disjoint. Apply the matrix to linear RGB, multiply by exposure and per-channel gains, clamp to $[0, 1]$, and round to uint16. The selected production recipe has no added sensor noise. These transforms increase variation, but do not reconstruct the original optics, sensor spectral sensitivities, or unknown processing already embedded in JPEGs and other source images.

Importantly, identical validation/test bytes across augmentation candidates do **not** mean unaugmented evaluation: both splits have their own fixed transformations, including held-out camera matrices. Only training payloads changed during augmentation selection. The released manifest freezes the actual transformation of every record.

The corpus contains 1,024,000 training records and 64,000 records in each evaluation split, for 1,152,000 total. TGPC is a data-only patch container: each 384-byte record includes a channel-major, little-endian uint16 7 by 7 RGB patch and provenance/augmentation fields. The 256-byte header binds the payload. The uncompressed file is 442,368,256 bytes. Deterministic gzip uses the existing zlib dependency; neither the trainer nor runtime requires a Python machine-learning library.

6 Native implementation

6.1 Algebraic specialization, not an additional likelihood approximation

For a d -dimensional observed marginal, set $C_k = \log \pi_k - \frac{1}{2} \log |S_k|$. With $\nu = 3$, S9 ranking is equivalent to ordering

$$\frac{\exp((C_k - C_{\max})/6)}{1 + \delta_k/3}.$$

For the full $d = 49$ stage and $T = 4$, weights are proportional to

$$\frac{\exp((C_k - C_{\max})/4)}{s_k^6 \sqrt{s_k}}, \quad s_k = 1 + \delta_k/3.$$

Model-only factors are precomputed. These identities remove per-pixel logarithms and exponentials without changing the mathematical K32/S9/q8 rule. Floating-point execution is still compared with tolerances, not assumed bit-identical across scalar, FMA, and different architectures.

6.2 SIMD organization and bounded work

The coarse AVX2 path evaluates eight components together for one target. After short-listing, requests are bucketed by component; full-stage vectors then process eight pixels sharing one expert, broadcasting its Cholesky coefficients and two conditional gain rows. Scalar tails handle incomplete groups. A scalar implementation and an ARM64/NEON implementation share the same contract. This is a CPU method; runtime GPU inference is not included.

OpenMP distributes phase/chunk work. Tiles schedule computation but do not create reconstruction boundaries: every neighborhood reads the complete input mosaic. Only the global boundary is reflected without edge repetition. The 512-pixel work chunk avoids allocating a full-frame 49-feature matrix. Recorded worker scratch is 419,840 bytes, about 7.2 MiB for eighteen workers, in addition to model, prepared coefficients, input, and RGB output storage.

6.3 RawTherapee and the model boundary

The clean implementation exposes “Student-t GMR (experimental)” with PP3 identifier `tgmr`. Markestijn three-pass remains the default. An installed data-only TGMR v2 model is authenticated and cached; an explicit `RT_XTRANS_TGMR_MODEL` override supports compatible custom models.

Runtime receives already preprocessed/scaled scalar RAW values. It does not apply a second white balance, a gamma wrapper, or an extra sharpening stage. Model training on rendered-image-derived RGB and runtime on camera data are different domains; matrix augmentation is only an approximation to that gap.

Missing/corrupt/incompatible models, unsupported CFAs, allocation failure, or non-finite inference trigger a diagnostic and complete Markestijn overwrite. This protects execution integrity. **It cannot detect a finite but visibly wrong interpolation, false color, or ringing.** Ordinary downstream profile processing can amplify differences and is not itself evidence of which demosaicer is colorimetrically correct.

7 Evaluation protocol and metric definitions

7.1 Three-channel center errors

All headline PSNR values below use the normalized range $[0, 1]$ and the complete center RGB vector, including the exactly restored measured channel:

$$\text{MSE}_{\text{RGB}} = \frac{1}{3N} \sum_{i=1}^N \sum_{c \in \{R, G, B\}} (\hat{x}_{i,c} - x_{i,c})^2, \quad \text{PSNR} = -10 \log_{10} \text{MSE}_{\text{RGB}}.$$

An earlier manuscript incorrectly described the denominator as two missing channels. The implementation and recorded numbers always used three. No result is rescaled in this revision. When the measured channel is exact, missing-channel-only PSNR is lower by $10 \log_{10}(3/2)$, approximately 1.761 dB. Comparisons require the same convention for every method.

Historical p99_abs means the 99th percentile of absolute scalar-channel errors, including the native-channel zeros. The production balanced-phase comparison instead reports the 99th percentile of center RGB RMS: $e_i = \sqrt{\sum_c (\hat{x}_{i,c} - x_{i,c})^2 / 3}$. These tail statistics are not interchangeable. The all-phase native validation report averages each record’s squared error over all eighteen phases before forming its patch RMS; that is a third, explicitly different aggregation from the balanced-phase comparison.

Per-source PSNR is computed from that source’s pooled errors; the paired source delta compares the same source under both methods. Median source delta is not the difference of two marginal median PSNRs. The production evaluator uses its recorded order-statistic convention for medians and p99.

7.2 Holdout boundaries and adaptive reuse

Development hyperparameters were selected on validation, and BSDS test, chromatic controls, Hubble/Hydra bright targets, and synthetic probes were excluded from parameter fitting. Nevertheless, failures on these controls motivated subsequent changes, notably Student-t stabilization. They are retrospective research diagnostics, not untouched confirmation of the entire adaptive design sequence. This distinction does not imply those images were inserted into the training patches.

For production-v1, augmentation and source-count studies used the frozen validation split. Two canonical fits established the model identity before the original test comparison. That comparison exposed hard cases and influenced retraining experiments; the old 500-source test is now correctly called diagnostic. No new independent confirmatory population is claimed.

7.3 Native production comparison

The production comparison reconstructs exactly 64,000 frozen centers from 500 sources. Each coordinate receives a deterministic balanced phase (record index modulo 18); it is not evaluated eighteen times in this particular head-to-head table. The separate trainer validation uses all eighteen phases.

Both demosaicers see the same transformed input and target. Markestijn uses 96-pixel source crops with padding, and a deterministic crop/full-image center check reported zero difference. That check supports this evaluated configuration, not a proof for every conceivable image and crop size. Source imagery is authenticated before decoding. The three-channel metric, phase assignment, crop geometry, and model identities are recorded alongside the results.

8 Results

8.1 Retrospective development evidence

The following table is generated from the frozen BSDS development records. It concerns research weights, not the later corpus-v1 model. GMAX is the earlier global joint spatial-chromatic linear estimator, not Markestijn.

Research configuration	RGB PSNR (dB)	p99 scalar absolute error
GMAX linear reference	32.054	—
Gaussian K64 experts and weights	34.835	0.084625
Gaussian experts, t weights ($\nu = 5$)	35.404	0.078690
Fitted dense K64 Student-t ($\nu = 3$)	35.518	0.077291
Reduced K32/S9/q8 ($\nu = 3$)	35.056	0.081494

The unchanged Gaussian-expert / Student-t-responsibility control isolates a useful effect of heavier-tailed weighting within that development setting. The fitted variant is stronger again, but also changes ν from five to three; that increment is not a fitting-only ablation. By contrast, differences between earlier full-RGB GMM and phase-conditioned GMR designs cannot be assigned solely to dimensionality because other settings changed too.

K32/S9/q8 trades some quality for arithmetic cost. The frozen reduction records about 90.8% retention of the dense K64 MSE gain over GMAX on these research test coordinates. Estimated MAC-scale operations fall from 163,072 to 23,264 per center. Such an arithmetic count is not a wall-clock prediction; memory access, shortlisting, square roots, and vector utilization matter.

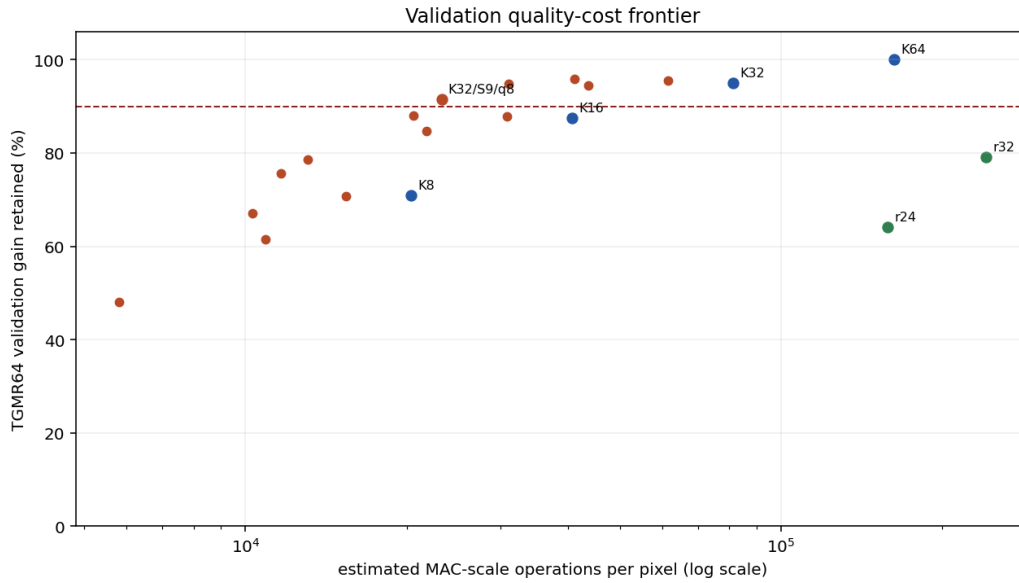


Figure 2: Historical validation quality-cost study. This selected the inference contract; it does not evaluate the production-v1 corpus.

The reduced research model improves some sparse-star results, but does not eliminate all sparse failures. The tiny-star control remains below GMAX. Uniform dark-background samples are insufficient evidence of star safety; the bright-target subsets must be reported separately.

Research subset	K32/S9/q8 PSNR (dB)	Delta to GMAX (dB)
Hubble bright targets	15.921	+0.486
Hydra bright targets	36.447	+3.663
Tiny-star control	8.267	-0.271

8.2 Production-v1: improvement with important exceptions

The original canonical balanced-phase population comparison is:

Diagnostic measure	TGMR production-v1	Markesteyn 3-pass
Pooled RGB PSNR (dB)	34.063575	32.316274
p99 center RGB RMS	0.095072	0.115206
Worst center RGB RMS	0.408201	0.483703
Median source PSNR (dB)	36.312600	34.701333

Pooled gain is **+1.747301 dB**; paired median source gain is **+1.625443 dB**. TGMR wins on **408** sources and loses on **92**. **65** sources lose more than 0.5 dB; the worst paired delta is **-6.169889 dB**.

Original gate	Required	Outcome
Pooled gain	at least +2 dB	FAIL
Median paired source gain	positive	PASS
Worst paired source loss	at most 0.5 dB	FAIL
Aggregate p99 RMS	no regression	PASS

Thus the candidate improves the population average and aggregate p99, but does not pass the originally required 2 dB pooled gain or the 0.5 dB maximum source-loss bound. Many source regressions are hidden by the aggregate gain. The gate decision is not changed merely because Markestijn also has visible errors in a difficult region.

The clean-branch repeat reproduced every TGMN statistic in the original comparison exactly. Its Markestijn pooled result was 32.317383 dB instead of 32.316274 dB. We retain the original table and its artifact identity above, rather than silently mixing baseline values from two builds. This small baseline difference does not change the failed-gate conclusion.

8.3 Signal strata

The production model improves pooled PSNR in each of the recorded marginal brightness, chroma, and texture strata. This is useful but weaker than safety on every source or every type of structure. The strata overlap, and their individual pixel counts must not be summed into additional independent data.

Marginal stratum	Centers	TGMN (dB)	Mark. (dB)	Delta (dB)
Brightness / low	23,611	48.560	45.888	+2.672
Brightness / middle	30,029	32.790	30.783	+2.007
Brightness / high	10,360	30.619	29.413	+1.206
Chroma / low	20,292	47.876	45.848	+2.028
Chroma / middle	20,273	35.425	33.561	+1.864
Chroma / high	23,435	30.919	29.214	+1.706
Texture / low	34,993	51.735	50.036	+1.698
Texture / middle	17,550	40.320	37.808	+2.512
Texture / high	11,457	26.928	25.239	+1.689

8.4 Hard-case retraining did not close the gap

Digitally sharp borders, very thin structures, and some isolated chromatic highlights can provoke false color or ringing. Such patterns are not representative of every photograph, but cannot be dismissed as impossible in actual camera data: labels, fine edges, saturated lights, and processing can supply related difficult structures.

A subsequent controlled screen replaced up to 5% of training patches with synthetic hard cases and compared direct extraction with a sensor-physical rendering variant. At 1,000

training sources, the best reported synthetic candidate reduced held-out synthetic mean MSE by only 15.83%, below the required 50%. Physical-rendering candidates lost 0.177–0.312 dB on ordinary validation. The screen therefore stopped without qualifying a full-scale replacement. The current model bytes were retained unchanged; neither the generator nor physical renderer is part of the clean inference product.

This is a negative result for the tested recipes, not proof that no training distribution could improve the failure. No new blind-test success or unexecuted full-factorial training is implied.

8.5 Numerical parity and timing

The research native-optimization study provides a controlled implementation measurement on a Ryzen 9 5900X with GCC 13.3, AVX2/FMA, and eighteen workers:

Research native measurement	Value
Reference/native error RMS	4.293898e-08
Maximum absolute error	4.768372e-07
Standalone mosaic pixels	39,959,032
Median mosaic time (s)	5.833451
Throughput (megapixels/s)	6.850
Measured center exact / thread-repeat deterministic	yes / yes

These are standalone mosaic measurements for the research model. They are not the execution time of the production trainer, a complete RAW export, or an ARM64 benchmark. The specialization agrees numerically with its reference; deterministic thread-count repetition does not imply that all scalar/FMA paths are bit-identical.

For the production model, clean-build descriptive full-RAW exports recorded 10.19–10.62 seconds and approximately 1.90 GiB peak RSS on the available X-T50 examples. Those runs occurred during other verification work and are not a controlled warm-up plus three-run performance gate. They establish successful execution, not a new portable speed claim. Full-frame portraits remain private; only reviewed earring crops accompany the repository.

9 Reproducibility and artifact identity

9.1 Corpus, trainer, and model

The public RT-TGMR-corpus release provides the compressed patch corpus, expanded coordinate metadata, attribution, reconstruction instructions, and checksums [13]. Original full photographs and private RAW exports are not release attachments. Reconstructing original images from third-party URLs is best effort; the authenticated TGPC payload is the durable training input. The release is a maintainer publication, not an official RawTherapee project release.

The canonical CPU trainer runs one thread with fixed ordering, float64 fitting, contraction disabled, and Ubuntu 24.04/GCC 13.3.0 as the pinned release environment. The fit uses seed 0x0000005847544d52, batches of 4,096, and an 8 GiB memory limit. Its configuration is serialized from actual checkpoint settings rather than an arbitrary supplied label. Source/build-input identity and corpus payload are bound to the model. The recorded original clean canonical fits produced byte-identical checkpoints and model artifacts; a later clean trainer build also re-exported the retained checkpoints to the same model. This manuscript rewrite does not constitute another training run.

Model size: **6,073,768 bytes**. SHA-256 values below are displayed as two consecutive halves; join them without whitespace.

Production-v1 TGMR v2 model

5707fbd67d1998ed3bac646ecce96729
7a2022776821d62944a24dbbb8615285

TGPC payload (not the compressed file)

573a0bf7f073282f715fbafc68ace1e8
366c8ca19e99e7f2f797a75c11444c74

Trainer configuration

13bf483df2104617785cd0c9be5eeb93
86a841a00b2e30087881811e7f19bcbf

Trainer source/build inputs

df0fc05b7b62f35c87a8ad42caf07e21
9721bddc81307cbf2756c9ba14d71ee6

Balanced-phase comparison JSON

9e2b799fa829d8cac33576a113594891
caf0396a1f30037f6164474aff8a6dba

The production artifact differs from the historical BSDS research model, whose SHA-256 is 6279b6a593ef4b595b1aff246182682b60c7eea701373ee2eb9f80cfcb50485c. Research and production numbers must never be associated with the other's coefficient file.

9.2 Evidence-linked document build

This paper is Markdown with a CSL YAML bibliography, rendered with pinned Pandoc 3.10.2 and citeproc to self-contained HTML and a typeset PDF. Title page and table of contents are retained. A verifier regenerates named result tables from authenticated tracked JSON, checks metric arithmetic and model identity, and rejects changed table text or inconsistent headline values. The companion README lists the evidence files and commands.

Git preserves earlier revisions. The document manifest authenticates this revision's inputs and rendered files; it is not a promise that every PDF rerender has identical binary bytes. PDF font subsetting can alter bytes without changing text or page appearance. HTML reproducibility and PDF text/layout stability are tested separately.

10 Limitations and release status

Not universal superiority. Production-v1 misses two important original quality gates. Its current status is an experimental option with accepted, documented limitations, not a fully qualified replacement for Markestijn. Finite-value/runtime checks do not provide a quality selector.

Adaptive research. Earlier test and external controls influenced the sequence of experiments. The current production test also became diagnostic after its failures motivated retraining. Independent confirmation remains necessary for strong generalization claims. No uncertainty intervals over independent source draws are supplied here, and the many phase views do not increase the number of independent images.

Approximate sensor domain. The training RGB derives from heterogeneous rendered images, including their earlier demosaicing, compression, and tone processing. Linearization and camera-matrix augmentation do not invert those operations. Noise, clipped highlights, extreme exposures, and camera-specific spectral effects can remain mismatched.

Confounding in comparisons. Development stages changed several design choices together. The paper does not isolate every contribution, reproduce all published competing systems, or establish a state-of-the-art ranking. Production and research datasets differ; their absolute PSNR values are not a before/after measure of the same evaluation.

Incomplete platform/camera matrix. GCC and Clang x86-64 checks and scalar/AVX2 parity support local viability. ARM64/NEON compilation is not native execution validation. Windows/macOS packaging, wider native hardware, and complete X-Trans generation I-V camera coverage remain qualification work. The currently documented X-T50 examples cannot stand in for that matrix.

Rights and publication boundaries. Corpus redistribution retains source attribution and individual terms. This does not itself finalize the trained model’s intended CC BY 4.0 licensing or every relevant personality right. Model-license finalization is separate from software licensing and from the already published corpus. A permanent archival DOI/second mirror is also distinct from the existing GitHub release.

11 Conclusion

Phase-conditioned Student-t regression provides a compact, interpretable statistical demosaicer with practical native CPU execution. Its specific advantages in this study come from modeling the measured task directly, using robust tempered component weighting, and organizing reduced inference around predictable local memory access. The downloadable corpus and authenticated model make the current result reproducible beyond the original research setup.

The evidence supports an experimental implementation, not an unconditional quality claim. The licensed diagnostic population improves on average, but some source losses are large and the tested hard-case retraining did not repair them. Retaining that distinction is part of the result: a useful average prior and efficient implementation can coexist with important, unresolved image-quality failures.

Reproducibility and assistance disclosure

Code, data identities, reports, and implementation references are associated with the pinned clean feature revision [14]. ChatGPT/Codex assisted with literature retrieval, code, experiments, and manuscript preparation. That assistance does not establish novelty, reference fidelity, or validity by itself. The artifact-linked checks support the specified numbers and contracts; broader conclusions remain bounded by the limitations above. This is a research/engineering manuscript, not a peer-reviewed paper.

References

- [1] M. Rafinazari and E. Dubois, “Demosaicking Algorithm for the Fujifilm X-Trans Color Filter Array,” in 2014 IEEE International Conference on Image Processing (ICIP), Oct. 2014, pp. 660–663. doi: [10.1109/ICIP.2014.7025132](https://doi.org/10.1109/ICIP.2014.7025132).
- [2] F. Markesteyn and RawTherapee contributors, Markesteyn X-Trans Demosaicing Implementation in RawTherapee. (Aug. 25, 2026). Available: https://github.com/RawTherapee/RawTherapee/blob/498f623784e33fd9a7077fcd8937fe0734033366/rtengine/xtrans_demosaic.cc
- [3] J. Portilla, D. Otaduy, and C. Dorronsoro, “Low-Complexity Linear Demosaicing Using Joint Spatial-Chromatic Image Statistics,” in IEEE International Conference on Image Processing 2005, 2005, pp. I-61–I-64. doi: [10.1109/ICIP.2005.1529687](https://doi.org/10.1109/ICIP.2005.1529687).
- [4] D. Kiku, Y. Monno, M. Tanaka, and M. Okutomi, “Beyond Color Difference - Residual Interpolation for Color Image Demosaicking,” IEEE Transactions on Image Processing, vol. 25, no. 3, pp. 1288–1300, Mar. 2016, doi: [10.1109/TIP.2016.2518082](https://doi.org/10.1109/TIP.2016.2518082).
- [5] M. Gharbi, G. Chaurasia, S. Paris, and F. Durand, “Deep Joint Demosaicking and Denoising,” ACM Transactions on Graphics, vol. 35, no. 6, pp. 1–12, Nov. 2016, doi: [10.1145/2980179.2982399](https://doi.org/10.1145/2980179.2982399).
- [6] F. Kokkinos and S. Lefkimmiatis, “Iterative Residual Network for Deep Joint Image Demosaicking and Denoising,” IEEE Transactions on Image Processing, vol. 28, no. 8, pp. 4177–4188, Aug. 2019, doi: [10.1109/TIP.2019.2905991](https://doi.org/10.1109/TIP.2019.2905991).
- [7] P. Sandeep and T. Jacob, “Joint Color Space GMMs for CFA Demosaicking,” IEEE Signal Processing Letters, vol. 26, no. 2, pp. 232–236, Feb. 2019, doi: [10.1109/LSP.2018.2886466](https://doi.org/10.1109/LSP.2018.2886466).
- [8] D. Zoran and Y. Weiss, “From Learning Models of Natural Image Patches to Whole Image Restoration,” in 2011 IEEE International Conference on Computer Vision, Nov. 2011, pp. 479–486. doi: [10.1109/ICCV.2011.6126278](https://doi.org/10.1109/ICCV.2011.6126278).
- [9] D. Peel and G. J. McLachlan, “Robust Mixture Modelling Using the t Distribution,” Statistics and Computing, vol. 10, no. 4, pp. 339–348, Oct. 2000, doi: [10.1023/A:1008981510081](https://doi.org/10.1023/A:1008981510081).
- [10] P. Ding, “On the Conditional Distribution of the Multivariate t Distribution,” The American Statistician, vol. 70, no. 3, pp. 293–295, 2016, doi: [10.1080/00031305.2016.1164756](https://doi.org/10.1080/00031305.2016.1164756).
- [11] F. Pedregosa et al., “Scikit-learn: Machine Learning in Python,” Journal of Machine Learning Research, vol. 12, pp. 2825–2830, 2011, Available: <https://jmlr.org/papers/v12/pedregosa11a.html>

- [12] P. Arbeláez, M. Maire, C. Fowlkes, and J. Malik, "Contour Detection and Hierarchical Image Segmentation," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 33, no. 5, pp. 898–916, May 2011, doi: [10.1109/TPAMI.2010.161](https://doi.org/10.1109/TPAMI.2010.161).
- [13] V. Haisman, "RT-TGMR Corpus: Reconstructible X-Trans Training Patches and Provenance." GitHub, Sep. 05, 2026. Available: <https://github.com/wilx/RT-TGMR-corpus/releases/tag/tgmr-corpus-v1>
- [14] RawTherapee contributors, RawTherapee. (Sep. 05, 2026). Available: <https://github.com/wilx/RawTherapee/tree/d49d20b080c51454ffe8e23db6bfd91504774466>